AI Data Centre Industry & Astera Labs – Chief Architect at a US-based Technology Firm

Consultant | 29 July 2025

Specialist Background

- Over 10 years' experience in the IT industry, focusing on AI and data centre strategy
- Well-placed to discuss technology debates around scale-up networking, as well as upcoming ASIC (application-specific integrated circuit) efforts and how they might look a little bit different from typical third-party merchant GPU architecture
- Knowledgeable on changes in the power distribution, as well as Astera Labs' retimer positioning and execution risk to get to the eventual 1,000-plus accelerators per pod

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If I think back to three years ago, I feel like the expectations were just as robust for CXL [compute express link]. The AI watershed moment came and everyone started focusing on training. To your point, CXL can become a lot more interesting for inferencing because inference workloads are more memory bottlenecks than compute or even networking bottlenecks. I see it as an emerging value proposition that's finally going to come into the forefront a little more because it seems like we're knocking on the door of inference CAPEX playing a larger role than training CAPEX, and the cost dynamics make plenty of sense to get 2TB of extra memory at maybe slightly lower latency for way cheaper. It's like USD 100-150 for some of these CXL controllers vs thousands of dollars for an incremental DIMM [dual in-line memory module]. The latency issue seems like the biggest bottleneck for CXL natively for the memory fabric. How do you see that being a problem? In certain inference use cases, latency sensitivity is of the utmost importance. Does that create some problems for CXL's value proposition?	13
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Power stages are super important, especially because we've got hundreds of different power stages in Blackwell, and that's only going to continue to increase with Rubin. I've also heard, though, that the ability to be able to sell both the controllers and the power stage is becoming increasingly important because at a point of failure, there are a lot of fingers at Monolithic Power, TI is going to point fingers at someone else and say that it's not the power-stage controller. There's a desire to have a single throat to choke. Is that true? If so, are any vendors missing one piece of the two-piece puzzle? For example, Onsemi's power stage is pretty decent, but they're still in the process of developing more quality controllers and maybe that will come. Maybe it will not be in time. Is that a dynamic that I should be aware of in your mind, that you'd want the ability for the vendor to be able to do both? Is there anyone on or otherwise that doesn't have that just now? 17

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Transcription begins

Analyst:
We've been talking a little bit about the scale-up networking dynamics, where NVLink has been very dominant, but it's a closed architecture. There's a handful of efforts to provide a quasi more open alternative to that, whether it's NV Fusion at the least open alternative, where you still have to take either an Nvidia CPU or GPU, but you can have the opportunity to introduce your own proprietary CPU or GPU alongside that and get the chip-to-chip interconnect. You've got UALink and then, as one-third, you've got scale-up Ethernet from Broadcom based on their Ultra Ethernet, Tomahawk switches. It sounds to me Amazon might be leaning in over the next 12-18 months or so more heavily into the X-Series Astera Labs PCA scale-up switch, which to me has always been the precursor to what eventually will be UALink. It seems like for the time being, because UALink isn't out yet, there's likely to be a decent ramp on this X-Series switch, but that there will be a continued debate around as UALink silicon comes into the market, whether or not you go with that long term or if you maybe opt for something like NV Fusion.

I haven't heard a ton of pickup on the scale-up Ethernet Broadcom version of things. You're RFP [request for proposal]-ing for 2027-28 demand, and you're looking at UALink, which doesn't have silicon out the gate yet, but has the X-Series from Astera Labs that you can spec against, and you're comparing it to NV Fusion and whatever is being offered there. I heard that specs are pretty sparse still. What would your main considerations as you weigh the choice between NV Fusion-based scalp-up networking and UALink-based scalp-up networking if you wanted to diversify away from NVLink in certain parts of your AI clusters?

Specialist:
That's a great question. I'd say, first of all, would be the bandwidth per GPU. I'd want to see exactly how much bandwidth I would get at the GPU level. I'd want to look at latency, making sure that it's definitely sub-microsecond. I'd want to look at the overall topology. Is it going to be something where it's like point-to-point where NV Fusion is that switch fabric or is it going to be UALink, which is mesh, can handle ring, can handle daisy chain, can handle switch fabric, things like that. I'd want to look at the memory semantics. Is it cash coherent like NV Fusion is or is it something like UALink is where it's memory load store semantics across a bunch of different heterogeneous accelerators, so I could not get locked in any one ecosystem. I'd want to look at the interconnect type. Is it something where like NVLink uses NVLink over proprietary SerDes vs UALink, which is really, frankly, modified Ethernet and uses custom transaction layers. I'd also want to look at what the maximum scale is because NV Fusion tops out at, what is it, 570 GPU, something like that, whereas UALink is 1,024 accelerators per pod.

With NV Fusion, you get better bandwidth, but you can have more accelerators in a pod with UALink. Other things I would look at too would be integration, looking at compatibility, what CPUs does it support, which GPUs does it support. With NV Fusion, it's only Nvidia GPUs vs UALink can use AMD, can use custom ASICs, can use Intel Gaudis, things like that and can use Nvidia ones. I want to look at the software stack because, honestly, ROCm vs CUDA, it's kind of a no concept there. ROCm has been getting much better. With UALink, you can also use SYCL and you can use custom drivers and things like that, too. With UALink though, I think Astera is a promoting founder of UALink as well. With UALink, you get buy-in from Azure, you get buy-in from Google, you get buy-in from Meta, things like that. NV Fusion is really tied to just the Nvidia ecosystem so you've got great software systems and DGX. I also want to look at what the, how to say this, strategic implications would be, when it comes down to Nvidia, and don't get me wrong, we're huge customers and partners with Nvidia. NVLink and NV Fusion really help Nvidia to maintain their dominance, whereas UALink, the idea is to democratise interconnects and make it so that it's easier to interconnect different hardware within your broader ecosystem.

UALink also is more about full openness across different vendors, whereas Nvidia is really about, you got to use Nvidia's hardware. You got to play in their CPU and GPU landscape. UALink is more about being consortium-driven and being collaborative vs something like NV Fusion where if Nvidia has issues because they are so proprietary, then you're tied to that, too. You've got NV Fusion out there now. Going forward, I think in 2027, probably 2028 is where you're going to see more commercial deployments of UALink and C-silicon out there. Honestly, UALink, I would say, is a pretty broad sell though to try to break Nvidia's monopoly on scale-up fabrics. It's not necessarily a bad thing, but Nvidia, they frankly dominated the market is what it comes down to. Whatever they say you're going to pay, you're going to pay. I think it's a good thing from a competition standpoint because it frankly forces them to innovate and be less proprietary.

Analyst:
On the dynamic around being able to scale up to more accelerators per pod, how much execution risk is there for Astera Labs to get to that eventual 1,000-plus accelerators per pod? My understanding is that the initial foray into the market will be at a much lower lane count-type switch, maybe in the 130-150 range. You're not going to get that full 1,000 accelerator scale-up until you start marching up towards closer to 500 lanes plus. How much execution risk do you think there happens to be or maybe not as it relates to being able to continue to build higher-rated switches, more lanes within the switch, given that they don't have a lot of switch know-how historically, and this is their big push into it? Are there any challenges that you're keeping your eyes on as it relates to their ability to actually move up to that full potential?

Specialist:
They've actually done a really good job of mitigating the risk. I would say interoperability, making sure that they've got seamless PCIe 6 and CXL 3 UALink compatibility across their GPUs, across their CPUs, across their switches. They've done that a lot with their Interop Lab. With the Scorpio switches, with the, what are they, Aries is the retimers and then the Leo controllers, they've done a really good job of making sure that interoperability between them that you'll be able to leverage different technologies from Nvidia, from AMD, from Intel, from Micron. I'd say signal integrity, that's always a challenge when it comes to those bigger pods, and you want to make sure that you've got better link stability, especially if you've got a long-reach topology or it's a very dense pod. With the Aries retimers and their gearboxes with optical modules that they've got, they really are built from a signal integrity where they're laser-focused on making sure that you've got as much stability as possible.

The other issue you typically have with those very dense pods is thermal management, power constraints and things like that. With the Cosmos software, they release a new version of Cosmos every other day. They keep optimising and optimising it for better routing, better for telemetry. They've got power aware diagnostics in there. With Cosmos and with the Scorpio switches, they've done a really good job of helping to manage any thermal dynamics or power constraints that you've got in there. I'd say that's the other thing, too, is Astera work very well with Nvidia, but at the same time, they are one of the founding members of UALink. When it comes down to it, they're the ones who help ratify the spec for 200G. They're the ones that actually set forth what the dynamics should be for having a 1,024 accelerator. They set the spec, they're setting their own bar. Then the other thing, too, is they're already shipping PCIe 6 components. They've got reference designs with Nvidia NGX. When it comes down to it, they're already at the foray of it. If anyone is probably poised to deliver on that amount of density, it would probably be Astera rather than some of the competition.

Analyst:
As you think about the need for UALink, you brought up the point that, from a strategic perspective, the way they maintain their dominance is. Would I be right to think that any type of UALink deployment is likely to be geared towards these non-Nvidia GPU-based clusters? Mainly ASIC [application-specific integrated circuit]-type deployments because Nvidia is probably going to close off any Nvidia-related opportunities. As you think about that ramping in the 2027-28 timeframe, that has something to do with the fact that that's when a lot of these custom ASICs are going to start ramping in, in more meaningful volumes?

Specialist:
Honestly, that's up to Nvidia and Jensen to decide that. It's not just his decision. If they're strategically going to try to lock you into their ecosystem, frankly, I think it'd be a bad move on their part because it's like when Intel had the dominance in the CPU market and they try to do some stuff like that, too. That gave rise to AMD and that gave rise to a lot of their competitors out there. UALink supports GPUs. It supports AMD. It supports Intel Gaudis. It supports other non-Nvidia GPUs and of course, supports ASICs, so if you want to do a Google GPU or an AWS Trainium, but you can also get chips from like Broadcom and Marvell and things like it, too. Also does support FPGAs, and it's not really a primary focus of it, but it does fully support that. It doesn't directly support Nvidia GPUs. You typically have to use NV Fusion. If anything, NV Fusion is going to be the outsider and if they want to continue to be proprietary and have you... That's the thing is they are the dominant GPU right now.

At the same time, interoperability. Look at what Linux did for cloud acceleration. We used to have with Windows. Windows was the dominant operating system. Then with Linux being open-source, the fact that you could actually just use it and it costs next to nothing, that's what made Linux the dominating operating system for the cloud, too. When it comes down to it, that whole walled-garden approach doesn't always work unless you've got such a big moat that you can crush your competitors. If anything, history has proven that that just doesn't work.

Analyst:
When the UALink spec first came out, I feel like there's a lot of industry debate that there wasn't a high-rated switch that was UALink compatible, and so that was going to have to get developed and people are saying that as a potential question mark because when you think about typical high-rated switches, there's only a handful of companies that do that such as Broadcom, Marvell and Cisco. It sounds like Astera Labs is taking more of the PCIe [peripheral component interconnect express]-native approach, where they've got a lot of know-how on the PCIe specification, and so that really lowers the execution bar for them because they're just going to make a pretty PCIe-like switch and don't need to focus too much on building a ground-up UALink one. Are there any shortfalls to taking that approach without looking at a UALink switch, from a ground-up perspective and taking a lot of what's already been developed on the PCIe switch side of things, using that and tweaking it a little feature set-wise to make sure that it's fully compatible with the UALink spec?

Specialist:
If anything, I think they're going to have a progression path. In terms of shortfalls, I don't know, it's a little nuanced is what it comes down to. If you're going to go to an all-in UALink-native fabric, then what it comes down to right now is that until hardware is going to be manufactured at scale, it's not

going to make much of a difference. With PCIe-based switches, there's definitely higher latency vs UALink, which are sub-microsecond. I would say with UALink, you definitely have full-load store, you've got atomic ops and things like that from a memory semantic standpoint, but PCIe is limited. With PCIe, you can only get to 64 accelerators vs UALink, you get the whole 1,024, and that's an ambitious goal, too. UALink is optimised for AI workloads and PCIe is general purpose. What it comes down to is they're basically setting their trajectory of making sure that the other thing too would be power and die where UALink has a leaner protocol stack vs PCIe, which has higher overhead. At the same time, PCIe is a little bit of a vendor lock-in is what it comes down to, whereas with UALink, they're able to basically say that it works with anything.

I would say the reason that they still build PCIe switches is that it's just ubiquitous among CPUs, GPUs, storage, things like that. It's really good for front-end connectivity. You get CXL memory expansion, too, which is great for inference workloads. I wouldn't expect them to go all in on UALink. I think that they'll probably still build PCIe switches in the future. Where the market is right now is it's PCIe, Generation 5, Generation 6 that they're already deployed. They're basically biding their time until UALink is more ubiquitous. Then I think that they're going to really switch over into that. Frankly, it's a dual strategy is what it comes down to is they want their Scorpio, their P-Series to help target like PCIe and CXL and then they're going to have Scorpio X, which will be more for UALink. It's really coming down to if you want to buy the switches, buy them for whichever technology is best in market. The problem with PCIe is that they're just not ultra-low latency. Right now, they're great but in two years, you're going to definitely want those X-Series switches.

Analyst:
How would you be thinking about your appetite to pay per lane for an X-Series or UALink switch? It sounds like people are resonating in the USD 5-7 per lane type of neighbourhood, which the 150 [sic] lane first version of the chip would suggest it's like USD 800-1,000 plus. Is that reasonable for you or is there a basis to think about what would drive appetite for dollar-per-lane type of ASPs for these types of products?

Specialist:
When it comes down to it, I would say you're in the right ballpark for any lane level silicon, I'd say it's a reasonable target, USD 5-7 per lane. If you're doing 200GB per second PHY and controller IP when it finally hits production, I think that really what it comes down to is... What Synopsys had like projections that it would probably be USD 5-7 per range. You also have to look at the switchboard overhead. You have to look at packaging, you have to look at retimers, which adds another USD 2-4 there. Your total bill of materials is going to be, before any level integrations can be anywhere from USD 7-11. That being said, because UALink has a much better protocol stack, it's going to reduce the die area, going to reduce the power and 31% of the cost of operating a data centre is power. It's a broader ecosystem, and it's an open ecosystem. You could use Broadcom, you could use Astera, you could use Synopsys, things like that. It's all interoperable IT. The pricing, just like supply and demand, is going to be more competitive.

I think you're also going to see just better scale-up efficiency. What is it the goal for UALink is to have 93% effective bandwidth at a much lower cost. From an economic standpoint, it's different trade-offs, too. It's the cost per accelerator, the radix trade-offs, the rack-level interconnects and things like that. USD 7-11 for a total bill of materials is probably more realistic.

Analyst:
When I think about how this fits into the rack ecosystem, the way that I understand it is typically, you'll have two raw GPU die per package, and then there'll probably be two ports sitting on the package, and you'll use one of those links for an X-Series or any sort of scale-up chip. You're looking at one switch chip per four GPU die is a reasonable way to think about how that might scale up in a

typical AI data centre. I'm sure there's some nuance to that. Is that relatively reasonable?

Specialist:
Two raw dies per package is one of the most common configurations, especially if you're using multi-chip modules and advanced packaging. I've seen single-die configurations where you've just got the monolithic chip. I've seen dual dies where you got two raw dies side by side or they can be stacked. I see multi-die where they've got several dies in just a single package with chiplets and stacked DRAM and HBM, which HBM is super hot right now. Stacked die where you get dies that are layered vertically completely. Then you get chiplet-based where there's modular dies with die-to-die interconnects, too. That's what AMD does with EPYC and Intel does with Ponte Vecchio and AI pods and things like that. There's a lot of different configurations in a package, frankly, two dies. The nice part about that, as you're alluding to, is that you can split the logic and split the I/O into separate dies, which ultimately gives you better wafer yield. From a process standpoint, you might have one die that's usually using advanced logic like 3nm and then the other more mature I/O, anywhere upwards of 28nm for process mixing.

The other benefit that gives you, frankly, is just thermal isolation. If you got a high-power compute from something that's very sensitive from an analogue standpoint, two dies does tend to be the better way to go. The other part that if you got one of those dies that fails or if you need to upgrade one of the dies, you can do so from a modularity standpoint without having to redesign the whole chip. That's why you see such a proliferation of the dual-die approach.

Analyst:
UAL spec number one came out, and then spec number two is going to come out later this year. I've been hearing that a big debate is that a lot of customers want optics to be included in the spec, maybe that's like PCIe over optics. I've heard that the main driver of that is that we've had the shift from passive DAC [direct attach copper] to active electrical, and there is a thought that the transition is going to continue to occur towards linear pluggable optics. There's going to be some sort of need for PCIe over optics compatibility for UALink as well. Is that something that you would expect to hear a lot more from over the next 6-12 months, that UALink becomes more PCIe over optics supportive and a lot of that is driven by that exact transition, AECs [active electrical cables] waning a little as optics reliability and costs improve over time?

Specialist:
All the talk and the chatter that I've heard about getting linear pluggable optics, LPOs in UALink 3, it really comes down to is the desire to scale AI clusters across racks. While at the same time, having low latency, having high bandwidth and then also being super energy efficient. Copper is limited to about four metres is what it comes down to, whereas optics are going to allow you to extend UALink beyond a single rack from a power standpoint, optical links are, frankly, a heck of a lot. They consume very little power is what it comes down to vs having to retime any electrical links. From a thermal standpoint, optics, they just don't have the heat density that a high radix switch environment is going to have. From a bandwidth scaling standpoint, any 200G lane optics that you have, they would actually do a really good job matching UALinks like PHY spec for future-proofing it. You'd be able to just go forward from a bandwidth standpoint and continue to scale it up. I'd also say just some of the pods that we're building, a lot of the pods that we used to build were 50-100 accelerators. Yes, as you're getting into the 1,000 accelerator range, any optics would enable that scale going beyond the four-metre rack-to-rack distance.

As we grow the size of the clusters, optics definitely play a bigger role in them, and that's because of the power, it's because of lower latency. Nice part about LPOs is they're hot swappable and they're super modular. Then from short reach standpoint, they are super cost-effective vs any of the other fabrics that you've got out there, too. CPOs at the same time, co-packaged optics, I would say they're looking to be more dominant from ultra-dense switch designs, but LPOs are frankly more flexible, you get a pluggable path for UALink too. The UALink consortium, they're more aligning with OIS and LPO MSA efforts, frankly, for ecosystem readiness. I would say, yes, you hit the nail on the head, LPOs are definitely going to play a role in probably 3.0.

Analyst:
I would think that Astera's retimer position is going to get attacked because of this. Thinking about retimers today, they're integrated right into the motherboard. The switch cost is a little higher because of that. That's already starting to come down a little because they do the smart cable modules without the retimers and DSPs, and the paddle chips that are in the actual cord. That's already pluggable in nature and more switch effective, and that will only increase as LPO [linear pluggable optics] starts to replace some of the interconnect dynamics. You mentioned it's probably more like USD 7-11 per lane, but that incremental dollar-per-lane content opportunity outside of the USD 5-7 that I mentioned for switch chips probably becomes more competitive for Astera because it's much more plug-and-play oriented and so you can attack the installed base a little more effectively that Astera has had so far on the retimer side of things as the signal integrity solutions, you become less integrated with the motherboard. Is that relatively reasonable as well?

Specialist:
Yes, definitely. I got to say that Astera has really done a really good job of diversifying their product portfolio. They've got smart retimers, they've got smart gearboxes. They've got fabric switches, what smart cable modules. They've got optical modules. They've got Ethernet modules. If anything, and they do CXL controllers, they do NVLink, they've done a really good job of instead of going all in on one technology, diversifying their product portfolio so that, yes, in the event that, let's say, retimers that you no longer need them integrated with LPOs, then you could still sell them, but because they've got optical modules and things like that that they'd be still well-positioned to take advantage of that progress where the industry is going.

Analyst:
On the LPO front, are Broadcom and Marvell, the main players to pay attention to that are doing the best work there? Are there any smaller-scale companies that you think I should keep my eyes on that might benefit a little from the shift to LPO?

Specialist:
I would say, yes, Broadcom have been pretty impressive. I would say Dell, they're looking at some end-to-end LPO deployments. Macom is another one. There's a company DustPhotonics. They do silicon photonics. Then there's an architectural firm called Latitude Design Systems. They're doing some pretty impressive stuff out there in terms of being thought leaders on LPO vs CPOs and deployment risk and how to consider them. Broadcom and Macom and then Dell would probably be the biggest ones.

Analyst:
On the CPO [co-packaged optics] front, you mentioned that it's going to be this corner case for ultra-dense switch deployments. I know there are still some challenges around high-loss radius in the instance of failure. If the laser goes down, then the entire switch might go down, and perhaps the entire GPU cluster goes down, gets solved by things such as 3D packaging or liquid cooling for the actual switch itself to eliminate some of the heat dissipation issues or do you think some of those high-loss radius failure issues are more sticky in nature?

Specialist:
Latitude. Just when it comes down to it, you want to make sure that you've got no single point of failure.

All of clusters and pods are designed to be doubly redundant. When you design it out like that, you can have single points of failure. As long as you've got layer dependencies, you've got the telemetry in there to help predict it. That way, you're able to really anticipate a failure before it occurs and then be more proactive about it. I think with liquid cooling, I think with single lasers as well, as long as you're designing for redundancy, then you can do a really good job. Each one of those clusters, every time we build one of those, those are USD 300m-400m per cluster.

Analyst:
Astera's competitors, Credo is one in particular that likes to knock Astera for their Cosmos, like saying that's not all that differentiated. I'm of the impression that it very much is, and it sounds like you are, too. From your perspective, what would be the key areas of differentiation that come from Cosmos that are super beneficial to you vs what you get offered from other vendors that are trying to take some socket share?

Specialist:
I don't know, Credo and I have a love-hate relationship.

Analyst:
You're probably talking to the same guy.

Specialist:
Yes I would say, Credo has always been more Ethernet-centric vs Astera. Credo has been around since 2008, whereas Astera, they were founded, what, 2016-17. They've really been geared towards more of the AI market and more being cloud-native. They don't think with that legacy mindset. Credo does optical DSPs, but they've been an Ethernet-based company. They do AECs and things like that vs Astera where they're really focused on AI fabrics, UALink, CXL, PCIe. They're both targeted towards hyperscalers, but where Astera leans heavily into the future with ASICs and with GPUs, Credo is more about enterprises and OEMs and the legacy data centre model. I would say that when it comes down to it, both of them are AI aligned, but I would say Astera does tend to have deeper integration with GPUs and with ASICs, whereas Credo is still strong in Ethernet, and they do PCIe Gen 6, but that's what they do with AECs, too. I would say Credo is still focused on the system-level design stack, whereas Astera with Cosmos, it's got excellent debugging.

There's a lot of different tools out there that come in the Cosmos stack as well. Credo, because they're so Ethernet heavy, it's weighing them down vs where Astera they're willing to be more... I don't know, Credo just tends to be more risk averse. How many companies would you see that build competing products within their four walls? You look at HP with the best jet and the laser jet printers. It's like, why would you create two products that compete against one another? It's like, there's two different markets. One is the home user and one is the office user. It's bold to find a company that's willing to do that and basically cannibalise itself. At the same time, that positions you better for the future so that if the market goes one direction, if it zig, you can zag and you're not caught off guard vs someone like Credo, which because it's so Ethernet-based and they lived in that Ethernet world that if the world does decide to go do away with Ethernet, then they're going to be left holding the bag there. Yes, we might be talking to the same people. Honestly, I don't know, Credo, they've got some challenges.

Analyst:
To the extent that you're pointing to the fact that Astera is very focused on diversifying its portfolio. The one area that I feel like they are behind in is the Ethernet AEC product. They've got the Taurus-branded AEC; it is mostly 400G today. They're talking about getting to 800G sometime this year. 800G has already been ramping, and there are paths to 1.6T and even conversations around 3.2T. When I think about the levels of priority for Astera, is Taurus something that they can succeed in as well or do

they have enough goodness going on elsewhere that I shouldn't expect them to start to gain market share in Ethernet-based AECs?

Specialist:
I would say, Credo is very good with Ethernet and with AECs. They dominate pretty much the market for any AECs or Ethernet retimers. They're optical DSPs, They've got a 3nm 200G lane DSP. They are really leading the forefront of getting out 1.6T optics. The challenge you also have with Credo though, is that they own the full stack. It's the SerDes IP, the retimers, the DSPs and the system design, which is a good thing because you do have fast innovation cycles, you don't have to worry about interoperability. When it comes down to it, I would say if you're looking towards building more AI training pods, scale-up fabrics and whatnot, Astera is just more aligned where things are going. If you're just looking to do retimer-heavy Ethernet fabrics, the sheer number of retimers we buy is insane. You know what a retimer is, the reason you have to have retimers is just none of the hardware is all patches where it's just basically like, this one is fast, this one is slow. Hey, we have to buy additional hardware to just retime it all to make sure that lane deployments are proper and that you've got that.

You can get down to not having to buy retimers, that's the ideal state where you don't need additional hardware to account for existing hardware. Do I think that Astera is going to get there? Yes, I'm actually a fan. I think that again, they lean into the future vs going, "We've got revenue coming from retimers so why would we want to do away with it?" They're basically like, "Look, if we can get rid of our retimer business and lean heavily into more interoperability, that's going to help us to sell many of our other products as well."

Analyst:
If I think back to three years ago, I feel like the expectations were just as robust for CXL [compute express link]. The AI watershed moment came and everyone started focusing on training. To your point, CXL can become a lot more interesting for inferencing because inference workloads are more memory bottlenecks than compute or even networking bottlenecks. I see it as an emerging value proposition that's finally going to come into the forefront a little more because it seems like we're knocking on the door of inference CAPEX playing a larger role than training CAPEX, and the cost dynamics make plenty of sense to get 2TB of extra memory at maybe slightly lower latency for way cheaper. It's like USD 100-150 for some of these CXL controllers vs thousands of dollars for an incremental DIMM [dual in-line memory module]. The latency issue seems like the biggest bottleneck for CXL natively for the memory fabric. How do you see that being a problem? In certain inference use cases, latency sensitivity is of the utmost importance. Does that create some problems for CXL's value proposition?

Specialist:
I honestly think with CXL Type 3 devices, they add DDR5 over PCIe. That you're basically getting around any of the CPU channel limits. I think CXL, frankly, from a latency standpoint, it really is designed to avoid traditional memory translation delays, which is actually really well positioned for inferencing. CXL also has that disaggregated architecture, too. You're able to do composable AI fabrics where memory and compute scale independently, which is really well aligned for inferencing. Then with CXL, you can do multiple accelerators. You can do GPUs and ASICs and they can share a unified memory pool. That way, you really reduce the fragmentation and you ultimately reduce latency too. If anything, CXL, from an ecosystem standpoint, it's definitely been emerging.

I think Astera and MemVerge and Supermicro, Micron too, and Samsung. They're all shipping CXL memory modules. From a performance standpoint, though, what is it, like 35-40% bandwidth boost and 24% speed up if you are using it for AI and HPC workloads. The software stack around it with Linux too, the Linux kernel 6.9 now supports doing what's called weighted interleaving, so you can do it across

DRAM and do it across CXL memory. If anything, CXL, to your point, is very well-positioned for inferencing, but also vector searching, LLM caching, RAG pipelines, things like that, where you don't need the speed that you would for a training workload, but inferencing because of memory disaggregation is perfect for that.

Analyst:
Thinking about the pooling opportunity, I've seen demonstrations where it's just like 8-16 Leo controller chips integrated onto a single PCB trace board. Is that how you'd think about the opportunity or is someone going to come out with a higher power box or integrated box that is more geared towards the pooling opportunity where you don't just get the 2TB of expanded capacity, you get much more on that?

Specialist:
I'm going to refrain from answering on that. I'm unable to answer that one.

Analyst:
When I think about insertion, originally, it was like talking about Granite Rapids and Turin; it seems like those are the 2.0 compatible CPUs. Should I wait until the 3.0 compatible CPUs come out, that that's when the CXL rise really starts to occur?

Specialist:
I don't know. I've seen some impressive stuff nowadays, too. I would say that 3.0 is definitely going to give it some benefit. I would say just when it comes to insertion hints and things like that, they are, frankly, the best way to optimise latency for that high-speed data transmission. When it comes down to it, I don't know, I think, frankly, you can always wait and wait and then lose the market opportunity, too. I would say, just to give it a couple more months, and you're going to see definitely some progress being made there.

Analyst:
They've mentioned that they've got some projects in flight that might have the opportunity to ramp towards the end of this year, so that aligns with how they've been talking about it, too. I think pooling is way more exciting than expansion. I wasn't sure to the extent that some of those early deployments might be more light volumes.

Let's think about any other issues with CXL. I know thinking about host biasing and some of the software layer elements of it are going to be important because of the composability and disaggregation of it. How are ecosystem efforts at the software layer one or two layers up from the actual controller chip itself, going? Are there still some challenges that need to be addressed?

Specialist:
I would definitely say, latency sensitivity is always an issue, and you get that with CRC and FEC fallback and things like that. You can mitigate that using different software with any NOP insertion complexity. It just comes down to how well you schedule it to maintain low latency flitting and things like that. If you do verification and debugging, then you can help to remedy that with CXL. Any memory coherency issues with CXL 2, if you've got like CXL.cache or CXL.mem interactions that can sometimes cause stale data or race conditions and things like that. Again, it comes down to your software stack and how well you control that too. With memory coherency, for example, so if you've got strict cache flushing policies and weighted interleaving and the Linux kernel, then it will take care of that. With latency tuning, you can use 256B latency optimised flip mode with NOP insertion hints, and it will address any of the CRC and the FEC transitions, some of the other issues you get.

The software stack with CXL in the past has been typically with the kernel, with the bias, with orchestration tools have been traditionally immature, but the software is honestly, as long as you validate that a CXL 3.0 compliant and use orchestration tools like MemMverge and things like that, then it's actually pretty good with handling it, too. From a security standpoint, that's always, you worry about shared memory pools. If any one of the operators in there poisons the pool or anything like that, then you can have an attack vector. With the security stuff, you can isolate the memory pools or you can enforce ACLs and monitor for any anomalous access patterns and stuff. The tools have been getting better out there. Then interoperability standpoint, not all CXL devices really identify the same. If you've got any consortium-certified devices and you run compliance tests across your different PCIe Gen 6 lanes, then you're pretty much guaranteed to be working. CXL's biggest strength is its flexible memory sharing is what it comes down to. You can use it for CPUs, GPUs, ASICs, anything, FPGAs, stuff like that. You just need to coordinate between the hardware, firmware, and the orchestration layers really well. The promise of CXL is playing out, but I would say that from an inferencing standpoint, it's really going to benefit.

Analyst:
From a SerDes front, 224G is the workhorse right now. I feel like there are debates around whether or not we see a jump to 400G-type SerDes or optics as we think about next-gen devices. What are the puts and takes as you think about the likelihood of whether or not 400G becomes the next workhorse SerDes or if SerDes starts getting attacked by optics over time?

Specialist:
These are great questions. You clearly have done your homework. This is one of the more technical interviews that I've ever had in quite some time. The coffee is still sitting in. With SerDes, if you want to move from 112G PAM4 to 224G PAM and things like that, that is what's going to give you 800G and 1.6T Ethernet with fewer lanes, smaller die, lower power, things like that. I've seen a lot of progress being made with LPOs. The big benefit of that is you're bypassing the SerDes bottleneck whatsoever. Frankly, you don't really have to worry about power or you don't have to worry about latency at scale, too. SerDes is like copper is what it comes down to it. SerDes over copper will definitely hit reach and power walls beyond anything four metres vs with LPOs. DSP-free optics is what it comes down to. They're 40% less power than a retime SerDes. There you have to power the retimer, too. I will say latency optics completely avoid retimer delays. They are definitely better for retiming and better for inferencing, sorry, and for memory coherence.

SerDes, honestly, its sweet spot is I'd say it's even less than four metres. It's more like two metres or less. AECs with SerDes are very cost-effective, very mature. Anything from a PCIe compatibility standpoint and with CXL compatibility, SerDes, it's very tightly coupled, but that's why it shines really well. If you look at going forward, I think you're going to see a hybrid approach where you can do PCIe Gen 6 over optics and you can combine SerDes PHYs with optical DSPs. UALink, though, is very optics friendly and using any Ethernet PHYs that support LPO or CPO modules out there. AI pods will still probably use SerDes going in the future, but only for intra-rack within that kind of two-metre stuff. Two metres, man, I can outstretch my arm and that's two metres right there, too. When it comes down to, that's not really far, especially within a rack. If you're going to top of the rack, the bottom of rack, that's five metres right there, four metres. That's why it just doesn't scale is what it comes down to.

Analyst:
For Astera Labs on the optical gearbox side of things, can you explain that? It's relatively new. I think it's a corner case for the time being. Can you explain where the optical gearbox comes into play? Is Astera doing anything overly differentiated on that front?

Specialist:

How to explain this? You seem pretty technical. I won't boil it down too much. If I had to explain it to my 15-year-old, then I'm sure she'd be like, "Dad, this is way over my head." It's basically a purpose-built silicon bridge that lets you do seamless connectivity between a PCIe 6 and a PCIe 5 device. The reason that's important is that some of our older PCIe 5 stuff, which isn't actually that old. It's like 1.5 years old. If you need to use it for AI and things like that, if you've got next-generation infrastructure, that way you don't have to rebuild the whole pod or rebuild the whole cluster, you can use next-generation infrastructure together. Doing protocol bridging between flip mode and between non-flip mode, but also having full bandwidth is really the benefit of that optical gearbox. The nice part there, too, is like I said, you're not doing a whole USD 350m investment. You can basically say, "Hey, we've got some older 1.5 years equipment. If you do PCIe over optics and things like that, you can have rack-to-rack and road-to-road GPU clustering." That's the big benefit of it. It avoids bottlenecks. It maximises your lane utilisation. You get disaggregated GPU technologies, you get even lower latency with optical interconnects.

Frankly, from a telemetry standpoint, you just get better uptime and fault isolation, because in the past, you had to go all PCIe 5 or all PCIe 6. Really, what's special about there is it's the first PCIe gearbox that's optimised for AI workloads, optical media and Cosmos, it's all software defined, too. You can do protocol state transition, you can do field upgrades, things like that. It actually has its own built-in analyser, too. You can do transmission lane margining, you can do error reporting, and it works across a whole bunch of different PCIe devices, too, so you're not really locked into their ecosystem. You can do it with open standards, too. Getting back, I described it as a bridge. It's not just a bridge. It's a smart optical aware interconnect fabric.

Analyst:
On the power distribution side of things, Nvidia is moving to this 800V HVDC architecture. It's going to be a separate power rack. To what extent are we going to see a pretty decent increase in wide bandgap products such as SiC [silicon carbide] and GaN [gallium nitride] vs pre-Rubin power distributions? Is it going to be meaningfully more non-silicon vs silicon-type power components or a gradual transition? Do players such as Navitas or on the SiC side of things such as Onsemi or Infineon, have the opportunity to take some incremental share from more silicon pure plays such as TI [Texas Instruments] or Monolithic Power, within the AI data centre?

Specialist:
It really is, honestly, HVDC, it is a radical rethinking of the way that you do, do architectures within the data centre, too. It's like a traditional 54V DC. Honestly, what it comes down to is you've got 200kW at the rack level. I've seen some of the specs of the 800V HVDC, they want to bring up to a full megawatt down to the rack level, which to me... I don't know, I've seen people lean on racks before and I'm like, "What are you doing? Don't lean on that." 1MW of power going through one rack, that would scare me. The big benefit of HVDC, though, is when it comes down to it, you're reducing the amount of copper you have for power distribution by up to 45%. From a 45% standpoint, you look at just the sheer amount of power that goes to racks, that is a huge cost savings there. From an efficiency standpoint, you don't have to have a whole bunch of transformers doing AC to DC conversion. That's the way that we traditionally do upgrades and whatnot are pretty incremental. This is more of a direct rid-to-rack conversion and then not having to do it.

Then from a cooling standpoint, any of those PSUs that you've got on a rack, those generate a lot of heat, and we've got to cool those down using localised cooling or liquid cooling and things like that. If you have centralised conversion, you're already inherently going to reduce the heat. If anything, frankly, from a semi standpoint, with GaN and with silicon carbide, the big benefit of those two is that GaN supports up to 650V and then silicon carbide honestly supports up to, I think it's like 6,500kV and things like that. You're going to see grid to 800V DC conversion, high-power rectifiers, things like that. With

GaN, you basically have secondary DC to DC conversion inside of the rack. That gives you higher switching speeds. It's going to give you definitely smaller, more efficient PSUs. You're going to have a lot lower conduction losses. We lose a lot of power and a lot of heat with conduction. Then frankly, from a longevity and a durability standpoint, you're going to have a longer lifespan, which means reduced maintenance because a lot of those AI clusters and things like that, we're rebuilding them every 1.5-2 years just to have to do upgrades and things like that.

If I can be guaranteed from a power standpoint that I won't have to rebuild and the equipment is much more durable, then yes, it's going to last longer, and we can see more ROI from it, too. Strategically, you're looking at scalability, you're looking at sustainability from a copper standpoint because copper is pretty expensive right now. Then from a scalability stuff with Rubin and the Rubin Ultra GPUs, you're going to have the ability to support full Kyber racks and things like that, too. A lot of what silicon carbide and GaN, they are designed for future-proofing, too, that once you get to 1MW and beyond racks, which scare me even more, that's where 2027-28, you're going to see 1GW-scale AI factories. That's what we're really trying to build towards, but still 1GW-scale building, that's a lot of power. Right now, the biggest bottleneck is still energy and the ability to get energy. You can design out megawatt scale racks and whatnot. If you don't have enough power going into the data centre, then that's still going to be the choking point.

Analyst:
Power stages are super important, especially because we've got hundreds of different power stages in Blackwell, and that's only going to continue to increase with Rubin. I've also heard, though, that the ability to be able to sell both the controllers and the power stage is becoming increasingly important because at a point of failure, there are a lot of fingers at Monolithic Power, TI is going to point fingers at someone else and say that it's not the power-stage controller. There's a desire to have a single throat to choke. Is that true? If so, are any vendors missing one piece of the two-piece puzzle? For example, Onsemi's power stage is pretty decent, but they're still in the process of developing more quality controllers and maybe that will come. Maybe it will not be in time. Is that a dynamic that I should be aware of in your mind, that you'd want the ability for the vendor to be able to do both? Is there anyone on or otherwise that doesn't have that just now?

Specialist:
I can't comment on individual vendors. You've got Flex, you've got Rohm, you've got Navitas. They're doing rack-integrated power shells and stuff like that and converters. Power stagers in general, they give you wide-bandgap semiconductors, which will help you with switch losses and have more high-efficiency converters, vertical power delivery, set systems, capacitive energy storage systems, things like that. When it comes down to, at all the different stages, going from the grid down to the individual chip, you've got the grid input, you've got substation conversion, you've got rack-level distribution, you've got any of the intermediate bus conversion point of load. As you are going from the grid directly to the chip, there's a lot of different power stages there. I think vendors, if they had more singular solutions, that's easy. Of course, they're more than happy to sell as many different transformers and all kinds of like boards and voltage regulators and things like that, too.

When it comes down to it, every time you go from one stage to another, you're going to have a loss is what it comes down to. With less loss, you would have less heat and less cooling costs. The energy we use is just obscene as it is. If you had, I would say, more modular stages that would allow you to have better, more flexible rack design and would support those 1MW densities, those would be ideal. Frankly, I haven't seen anyone that's really doing it. I've seen plans for it, but I haven't seen anyone that's doing it just now.

Analyst:

There's been a lot of fast-moving developments, and I'm sure it's not going to slow down too much.

Transcription ends